

Financial Advisor Project - Analysis Report

1. Project Overview

The Financial Advisor project is a full-stack AI-powered financial advisory platform. It provides users with real-time stock metrics, financial news, portfolio management, and an AI-driven chatbot for financial queries. The platform uses a React-based Next.js frontend and a Python-based Flask backend.

2. Architecture & Tech Stack

- Frontend: Next.js, React, Tailwind CSS, Recharts, Chart.js, Framer Motion.
- Backend: Python, Flask, Flask-CORS, Flask-JWT-Extended.
- Database: MongoDB (via PyMongo).
- External APIs: yfinance (Market Data), Alpha Vantage, Google Generative AI (Gemini), News APIs.

3. Key Features

1. User Authentication: Secure login and registration using JWT.
2. Dashboard: A comprehensive view of market trends, global and Indian financial news.
3. Stock Metrics: Live financial data analysis and visualization.
4. Portfolio Management: Users can track and manage their investments.
5. AI Chatbot: An integrated Gemini-powered assistant capable of answering financial questions.

4. Running the Project

Backend (Flask):

1. Create a virtual environment and activate it.
2. Install dependencies via ``pip install -r backend/requirements.txt``.
3. Add a ``.env`` file in the backend folder containing ``MONGODB_URI``, ``JWT_SECRET_KEY``, etc.
4. Run the backend server using ``python app.py`` on port 5000.

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Frontend (Next.js):

1. Navigate to the root directory.
2. Install Node dependencies with ``npm install``.
3. Start the development server using ``npm run dev`` on port 3000.